DECLARATION

I, Ayush Sharma, hereby declare that the project titled *"Bonita GPT: A Voice-Enabled AI Virtual Assistant with Graphical User Interface"*, submitted for the partial fulfilment of the degree of Bachelor of Computer Applications, is my original work and has been carried out by me under proper guidance and supervision. This project represents my sincere efforts in understanding and applying the concepts of Artificial Intelligence, voice recognition, and GUI-based application development using Python.

I further declare that this work has not been submitted, either wholly or partially, to any other institution or university for the award of any degree, diploma, or certification. All the information, ideas, and materials used in this project have been properly acknowledged wherever necessary.

This project has been developed by me as part of my academic curriculum, and it reflects my knowledge, skills, and learning acquired during the course of study. I take full responsibility for the authenticity and accuracy of the content presented in this report.

I also confirm that all the work presented in this project report is free from plagiarism and has been completed honestly and ethically. Any references used have been appropriately cited in the references section.

CERTIFICATE

This is to certify that the project report entitled *"Bonita GPT: A Voice-Enabled AI Virtual Assistant with Graphical User Interface"* submitted by **Ayush Sharma**, a student of BCA [6th Semester], is a record of original work carried out by him under my supervision.

This work has been completed in partial fulfillment of the requirements for the award of the degree of Bachelor of Computer Applications (BCA) from GGDSD College, Chandigarh.

To the best of my knowledge, this project has not been submitted elsewhere for the award of any other degree or diploma.

Supervisor's Signature: _____

ACKNOWLEDGEMENT

I, Ayush Sharma, express my sincere gratitude to Dr. Monika Sethi Ma'am for her continuous support, valuable guidance, and constant encouragement throughout the development of this project. Her insightful suggestions and expert knowledge helped me to understand the concepts more clearly and improve both technically and academically. Her motivation played a significant role in the successful completion of this project.

I am also thankful to my institution, GGDSD College, Chandigarh, for providing the necessary resources, facilities, and a supportive learning environment that enabled me to carry out this project effectively. The availability of proper infrastructure and academic support contributed greatly to the smooth execution of my work.

I would like to extend my appreciation to all the faculty members who have guided me during my academic journey and helped me build a strong foundation in computer science and programming concepts.

Lastly, I would like to thank my family and friends for their constant encouragement, moral support, and motivation throughout the course of this project. Their support helped me stay focused and complete this work successfully.

ABSTRACT

“Bonita GPT: A Voice-Enabled AI Virtual Assistant with Graphical User Interface” is a desktop-based application developed using the Python programming language. The project aims to demonstrate the integration of Artificial Intelligence, voice recognition, text-to-speech processing, and graphical user interface design into a single interactive system.

The application is designed to function as a smart virtual assistant that can accept both text and voice input from the user and generate intelligent responses using an AI model. The system utilizes speech recognition technology to convert spoken input into text and employs text-to-speech functionality to deliver responses in an audible format, thereby enhancing user interaction and accessibility.

The assistant is capable of performing various tasks such as answering user queries, providing real-time information like date and time, and executing basic system commands such as opening web applications (YouTube, Google, WhatsApp) and local programs. The integration of a conversational AI model enables the system to provide meaningful and context-aware responses.

The graphical user interface is developed using CustomTkinter, providing an interactive and user-friendly environment. The interface includes features such as a chat display panel, input field, control buttons, and animated visuals to improve the overall user experience.

The project follows a modular programming approach, where different functionalities such as AI interaction, voice processing, command handling, and GUI management are implemented as separate components. This improves the maintainability, scalability, and efficiency of the application.

Overall, this project provides practical exposure to real-time AI application development and demonstrates how multiple technologies can be combined to create an intelligent and user-friendly virtual assistant system.

LITERATURE REVIEW

The development of intelligent virtual assistants has gained significant attention in recent years due to rapid advancements in Artificial Intelligence (AI), Natural Language Processing (NLP), and Human-Computer Interaction (HCI). Virtual assistants such as Siri, Google Assistant, and Alexa have demonstrated how AI can be integrated into everyday applications to improve user convenience and productivity. These systems rely on speech recognition, language understanding, and response generation to interact effectively with users.

Several research studies have focused on improving the accuracy and efficiency of speech recognition systems. Modern speech recognition technologies convert spoken language into text using machine learning algorithms and large datasets. Libraries such as SpeechRecognition in Python have made it easier for developers to implement voice-based input systems in applications. However, challenges such as background noise, accent variations, and speech clarity still affect performance.

Text-to-speech (TTS) systems are another important component of virtual assistants. These systems convert textual responses into audible speech, enabling a more natural interaction between humans and machines. Tools like pyttsx3 provide offline speech synthesis capabilities, allowing developers to integrate voice output without relying on internet connectivity. Research in this field focuses on improving voice naturalness, pronunciation accuracy, and emotional tone.

Conversational AI models have evolved significantly with the introduction of advanced language models. These models are capable of understanding context, generating meaningful responses, and maintaining conversational flow. APIs such as Google Generative AI (Gemini) allow developers to integrate powerful AI capabilities into their applications, making it possible to build intelligent chat-based systems with minimal complexity.

Graphical User Interface (GUI) design also plays a crucial role in enhancing user experience. Frameworks like Tkinter and CustomTkinter enable the creation of interactive and visually appealing interfaces. Studies show that a well-designed GUI improves usability, accessibility, and overall user satisfaction. The inclusion of features such as chat panels, buttons, and animations further enhances interaction.

This project builds upon these existing technologies by integrating speech recognition, text-to-speech, conversational AI, and GUI design into a single system. The aim is to develop a user-friendly virtual assistant that demonstrates the practical implementation of these concepts in a real-time application environment.

INTRODUCTION

"Bonita GPT: A Voice-Enabled AI Virtual Assistant" is an interactive desktop application designed to simulate real-time human-computer interaction using Artificial Intelligence. The system enables users to communicate through both text and voice, providing intelligent responses and performing various tasks efficiently.

The application integrates multiple modules such as:

- User input system (text and voice-based interaction)
- AI response generation system
- Speech recognition system
- Text-to-speech system
- Graphical User Interface (GUI) system

The working of the system is based on real-time processing, where user input is analyzed instantly and appropriate responses are generated. The assistant is capable of performing tasks like answering queries, opening web applications, and providing real-time information such as date and time, making it a practical and user-friendly solution.

OBJECTIVES OF THE PROJECT

The main objective of the *Bonita GPT: Voice-Enabled AI Virtual Assistant* project is to design and develop an intelligent system that enhances human-computer interaction through the use of Artificial Intelligence and voice-based technologies. The project aims to provide a smart, interactive, and user-friendly assistant capable of understanding user input and delivering meaningful responses.

The specific objectives of the project are as follows:

• To develop an AI-based virtual assistant that can understand and respond to user queries using natural language.

• To integrate speech recognition technology for converting voice input into text, enabling hands-free interaction.

• To implement text-to-speech functionality so that the system can provide audible responses to users.

• To design a user-friendly graphical interface using CustomTkinter for smooth and interactive communication.

• To enable the assistant to perform basic system operations such as opening websites and applications.

• To provide real-time information such as current date and time through the assistant.

• To develop the system using a modular programming approach for better maintainability and scalability.

• To enhance user experience by combining visual interface, voice interaction, and intelligent response generation.

• To gain practical knowledge and hands-on experience in integrating Artificial Intelligence with real-time applications.

Overall, the project aims to demonstrate how modern AI technologies can be effectively used to create a smart and efficient virtual assistant system.

KEY FEATURES

The *Bonita GPT: Voice-Enabled AI Virtual Assistant* offers a variety of features that make it an intelligent, interactive, and user-friendly system. The key features of the project are as follows:

• **Dual Input System (Text and Voice):**
The application allows users to interact using both typed text and voice commands, making it flexible and accessible.

• **Speech Recognition:**
The system uses speech recognition technology to convert spoken input into text for further processing.

• **Text-to-Speech Output:**
The assistant responds not only in text form but also speaks the output, enhancing user interaction and accessibility.

• **AI-Powered Responses:**
The application generates intelligent and context-aware responses using an AI model, enabling natural conversation with the user.

• **Real-Time Processing:**
User inputs are processed instantly, and responses are generated without noticeable delay, ensuring smooth interaction.

• **Graphical User Interface (GUI):**
A modern and attractive interface is designed using CustomTkinter, including chat display, input field, buttons, and animations.

• **Command Execution:**
The assistant can perform predefined tasks such as opening websites (YouTube, Google, WhatsApp) and launching system applications like Notepad.

• **Time and Date Information:**
The system can provide real-time information such as current time and date upon user request.

• **Interactive Chat Display:**
All conversations between the user and the assistant are displayed in a structured chat format for better readability.

• **Voice Control Features:**
The system includes commands like “stop” and “resume” to control speech output dynamically.

• **Error Handling:**
The application handles errors such as microphone issues or network failures and provides appropriate feedback to the user.

• **Modular Design:**
The project is developed using a modular approach, making it easy to maintain, update, and expand in the future.

Overall, these features make Bonita GPT a powerful and efficient virtual assistant that enhances user experience and demonstrates the practical implementation of Artificial Intelligence.

FEASIBILITY STUDY

The *Bonita GPT: Voice-Enabled AI Virtual Assistant* project is feasible in terms of technology, cost, and usability.

- **Technical Feasibility:**
  The system is developed using Python and easily available libraries, making it simple to implement on standard computers.
- **Economic Feasibility:**
  The project is cost-effective as it uses free and open-source tools, requiring no expensive resources.
- **Operational Feasibility:**
  The application is user-friendly and allows interaction through text and voice, making it easy to use for all users.
- **Time Feasibility:**
  The project can be completed within the given academic timeframe due to its modular design.

Overall, the project is practical, efficient, and suitable for real-world use.

HARDWARE REQUIREMENTS

The *Bonita GPT: Voice-Enabled AI Virtual Assistant* does not require high-end hardware and can run efficiently on a standard computer system. The basic hardware requirements are as follows:

- **Processor:** Intel Core i3 or higher
- **RAM:** Minimum 4 GB (8 GB recommended for better performance)
- **Storage:** At least 500 MB of free disk space
- **Input Devices:** Keyboard and Mouse
- **Microphone:** Required for voice input functionality
- **Speakers/Headphones:** Required for audio output (text-to-speech)
- **Display:** Standard monitor with minimum resolution of 1024×768

These hardware components are sufficient to ensure smooth functioning of the application and provide a good user experience.

SOFTWARE REQUIREMENTS

S. No.	Software / Tool	Description
1.	Python	Programming language used to develop the application
2.	Google Generative AI API	Used to generate intelligent and conversational responses
3.	SpeechRecognition Library	Converts voice input into text
4.	pyttsx3	Converts text output into speech
5.	CustomTkinter	Used to design the graphical user interface (GUI)
6.	PIL (Pillow)	Handles images and animations in the interface
7.	Web Browser	Required to open websites like YouTube, Google, and WhatsApp
8.	Operating System (Windows)	Platform required to run the application
9.	Code Editor (VS Code / IDLE)	Used for writing and executing the program

SYSTEM MODULES

1. User Input Module

The User Input Module is responsible for handling both text and voice inputs from the user. It captures user queries through the keyboard or microphone and forwards them to the system for processing.

Input / Output

Input	Output
Text / Voice Input	Processed Text Input

Data / Attributes

- User Text
- Voice Input
- Converted Text

Process Flow

ENTER INPUT → CONVERT (if voice) → SEND TO SYSTEM

2. AI Processing Module

This module is the core of the system. It processes the user input and generates intelligent, context-aware responses using the AI model.

Code (Conceptual)

```
response = chat.send_message(user_input).text
```

Flow

INPUT → AI MODEL → GENERATE RESPONSE

3. Command Handling Module

This module checks whether the input is a predefined command. If matched, it executes the command; otherwise, it forwards the input to the AI module.

Examples of Commands

- Open YouTube
- Open Google
- Show Time / Date

Code

```
if "open youtube" in command:
    webbrowser.open("https://www.youtube.com")
```

Flow

INPUT → CHECK COMMAND →
MATCH → EXECUTE
NOT MATCH → SEND TO AI

4. Speech Module (Text-to-Speech & Speech Recognition)

This module handles voice interaction. It converts speech to text and text to speech for communication.

Data / Attributes

- Audio Input
- Text Output
- Voice Output

Code

```
engine.say(text)
engine.runAndWait()
```

Flow

VOICE → TEXT → PROCESS → SPEECH OUTPUT

5. GUI Module

The GUI Module manages the graphical interface using CustomTkinter. It displays chat messages, buttons, input fields, and animations.

Features

- Chat Window
- Input Field
- Buttons (Send / Speak)
- Animation

Flow
USER ACTION → GUI → DISPLAY OUTPUT

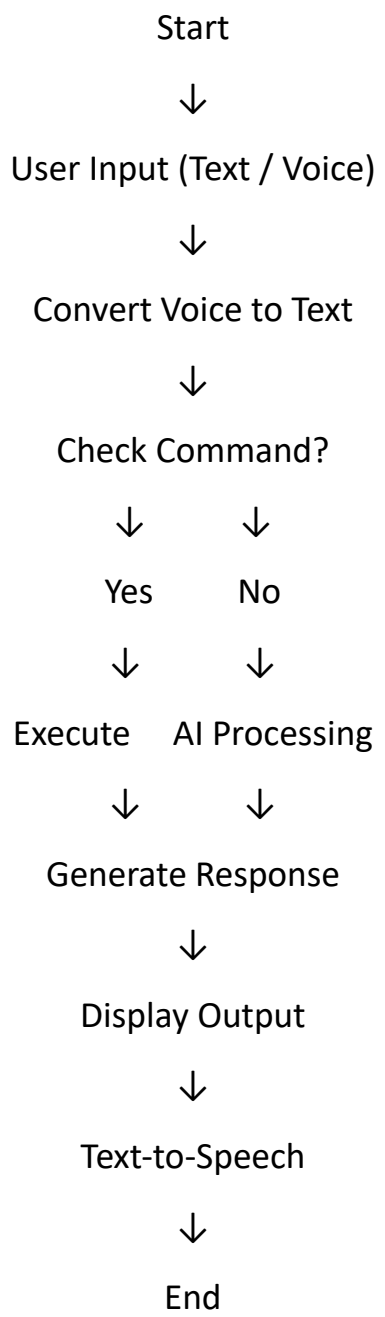
---

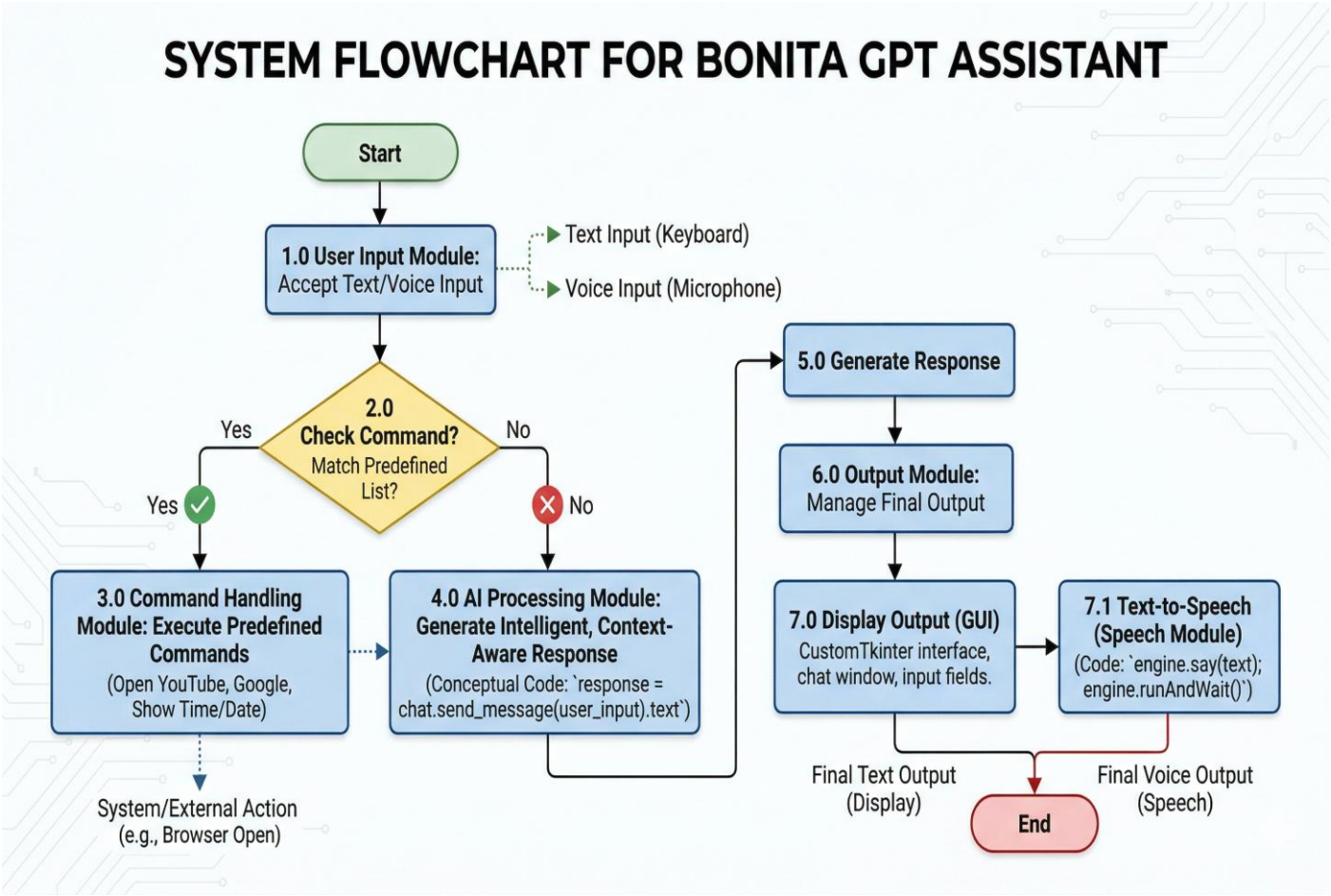
6. Output Module

This module displays the final response in text form and also sends it to the speech module for audio output.

Flow
GENERATE RESPONSE → DISPLAY → SPEAK

SYSTEM FLOWCHART





MODULES TABLE

S. No.	Module Name	Description
1.	User Input Module	Accepts text and voice input
2.	AI Processing Module	Generates intelligent responses
3.	Command Handling Module	Executes predefined commands
4.	Speech Module	Handles speech recognition and text-to-speech
5.	GUI Module	Manages graphical user interface
6.	Output Module	Displays and speaks the final response

Overall, the system modules work together to provide a smooth, intelligent, and interactive user experience. The modular design ensures easy maintenance and future enhancements.

FUNCTION LOGICS

The *Bonita GPT: Voice-Enabled AI Virtual Assistant* operates through a set of well-defined functions, each responsible for a specific task. These functions work together to ensure smooth execution of the system.

---

1. speak() Function

This function converts text into speech using the text-to-speech engine.

Logic:
- Takes text as input
- Splits long text into smaller sentences
- Speaks limited sentences for better clarity
- Uses engine to produce voice output

Code

```
def speak(text):
    engine.say(text)
    engine.runAndWait()
```

---

2. handle_commands() Function

This function checks if the user input matches predefined commands and executes them.

Logic:
- Convert input to lowercase
- Check for keywords (YouTube, Google, etc.)
- Execute corresponding action
- Return response message

Code

```
if "open youtube" in command:
    webbrowser.open("https://www.youtube.com")
```

---

3. add_chat() Function

This function displays messages in the chat window.

Logic:
- Takes sender name and message
- Aligns text (left/right)
- Adds message to GUI frame

---

4. get_response() Function

This is the main function that processes user input and generates responses.

Logic:
- Get input from entry field
- Display user message
- Check command
- If command → execute
- Else → send to AI model
- Display response
- Call speak()

Code

```
response = chat.send_message(user_input).text
```

---

5. get_voice_input() Function

This function captures voice input from the user.

Logic:
- Activate microphone
- Listen to user voice
- Convert speech to text
- Insert text into input field
- Call response function

---

6. update_gif() Function

This function updates animation frames in GUI.

Logic:
• Loop through image frames
• Change image continuously
• Use timer for smooth animation

---

7. auto_greet() Function

This function greets the user when the application starts.

Logic:
• Display welcome message
• Call speak() to greet user

FUNCTION FLOW

Start

↓

User Input

↓

get_response()

↓

Check Command → handle_commands()

↓

If No → AI Processing

↓

Display → add_chat()

↓

Speak → speak()

↓

End

Overall, these function logics ensure that the system works efficiently by dividing tasks into smaller, manageable units, improving performance and maintainability.

WORKING OF THE PROJECT

The *Bonita GPT: Voice-Enabled AI Virtual Assistant* works by integrating multiple modules such as user input, AI processing, command handling, and output generation to provide a smooth and interactive experience. The system operates in real-time and follows a step-by-step process as described below:

Step 1: User Input

The process begins when the user provides input to the system. The input can be given in two ways:

- Text input through the keyboard
- Voice input through the microphone

If the input is voice-based, it is first converted into text using speech recognition technology.

Step 2: Input Processing

The system receives the input text and processes it by converting it into a suitable format. It also removes unnecessary spaces and prepares the query for further analysis.

Step 3: Command Checking

The system checks whether the user input matches any predefined commands such as:

- Opening websites (YouTube, Google, WhatsApp)
- Opening applications (Notepad)
- Providing date and time

If a command is detected, it is executed immediately.

Step 4: AI Response Generation

If the input is not a predefined command, it is sent to the AI model. The AI processes the query using natural language understanding and generates a relevant and meaningful response.

Step 5: Output Display

The generated response is displayed on the graphical user interface in the chat window. The system maintains a proper conversation format between the user and the assistant.

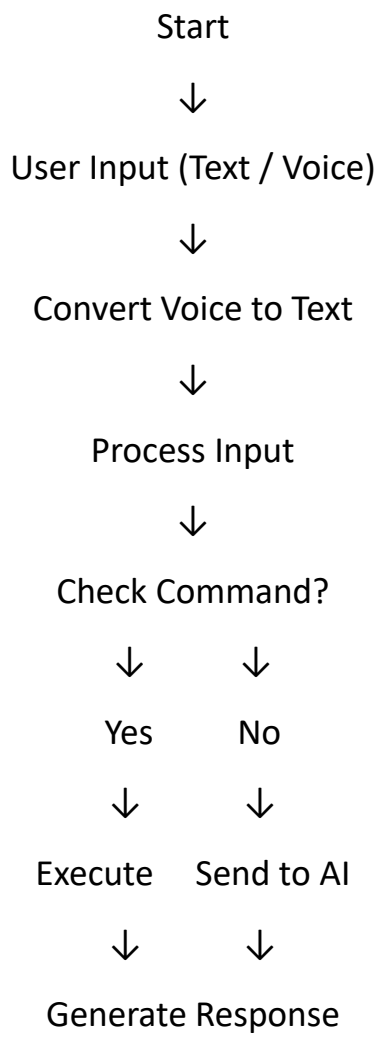
Step 6: Voice Output

The response is then converted into speech using the text-to-speech module. This allows the assistant to speak the response, making the interaction more natural and user-friendly.

Step 7: Continuous Interaction

The system continues to run in a loop, allowing the user to interact multiple times. It waits for the next input and repeats the process.

WORKING FLOW



↓

Display Output

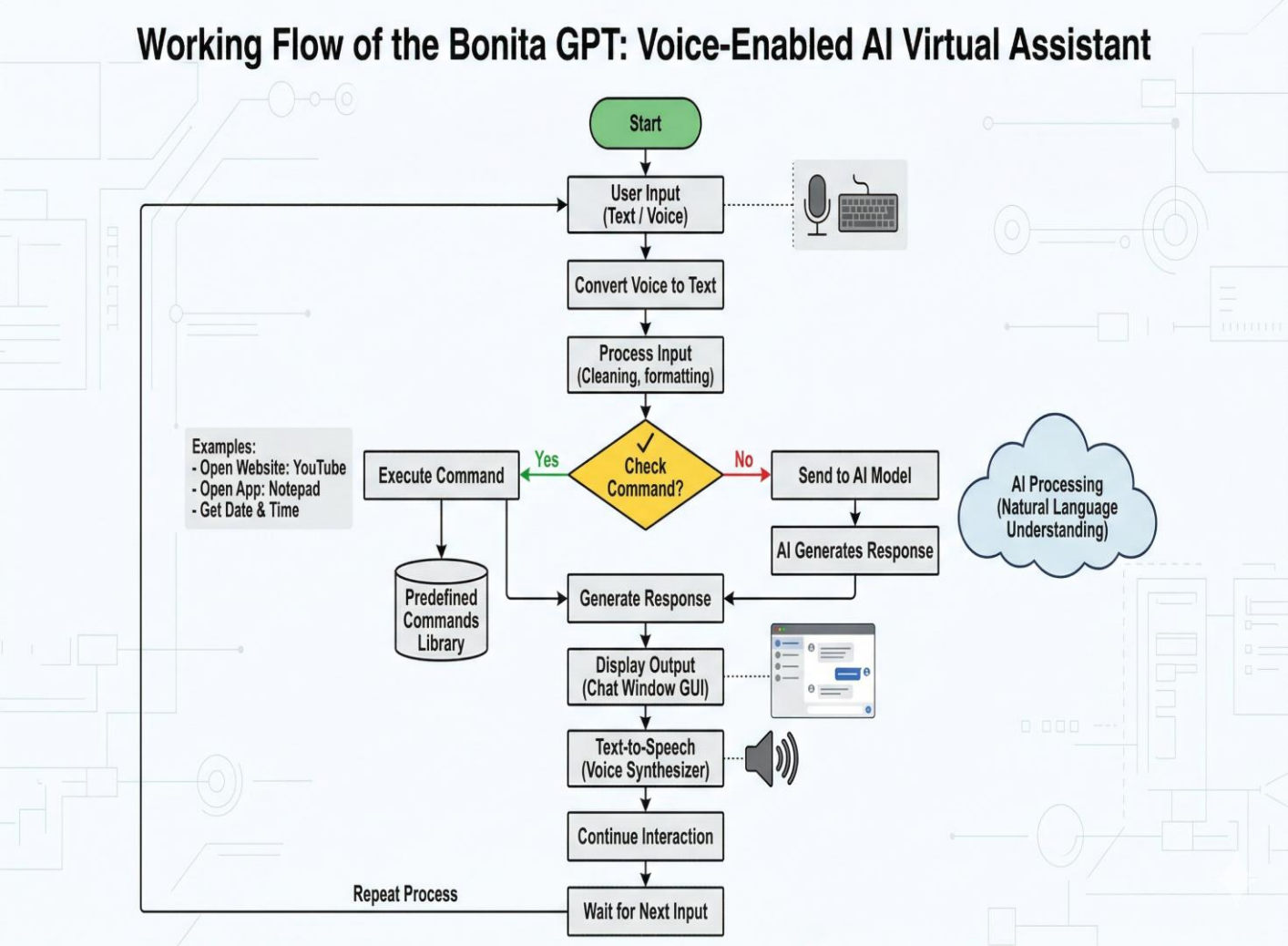
↓

Text-to-Speech

↓

Repeat Process

Overall, the working of the system ensures smooth communication between the user and the assistant by combining AI, voice processing, and GUI technologies in a real-time environment.

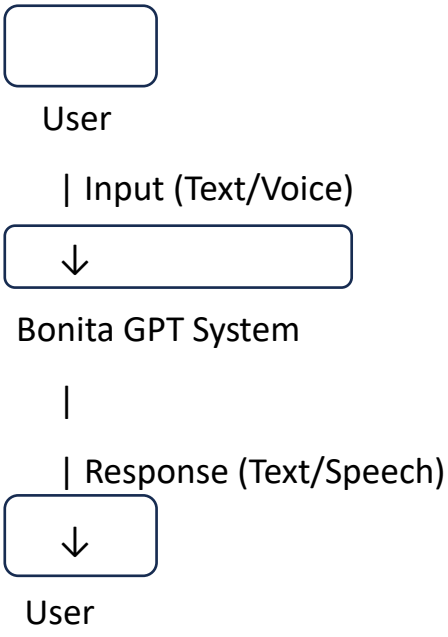


DATA FLOW DIAGRAM (DFD)

A Data Flow Diagram (DFD) represents how data moves through the system and how different modules interact with each other. It helps in understanding the overall structure and working of the system.

DFD Level 0 (Context Diagram)

This is the highest-level diagram that shows the interaction between the user and the system.



DFD Level 1

This diagram shows the internal working modules of the system.

User

↓

[User Input Module]

↓

[Speech Recognition] (if voice)

↓

[Command Handling Module] → (Execute Commands)

↓

[AI Processing Module]

↓

[Output Module]

↓

[Text-to-Speech Module]

↓

User

DFD Level 2 (Detailed View)

This diagram provides a more detailed representation of the system processes.

User Input (Text/Voice)

↓

Convert Voice to Text

↓

Process Input

↓

Check Command?

↓ ↓

Yes No

↓ ↓

Execute AI Model

Command ↓

↓ Generate Response

↓

Display Output (GUI)

↓

Convert to Speech

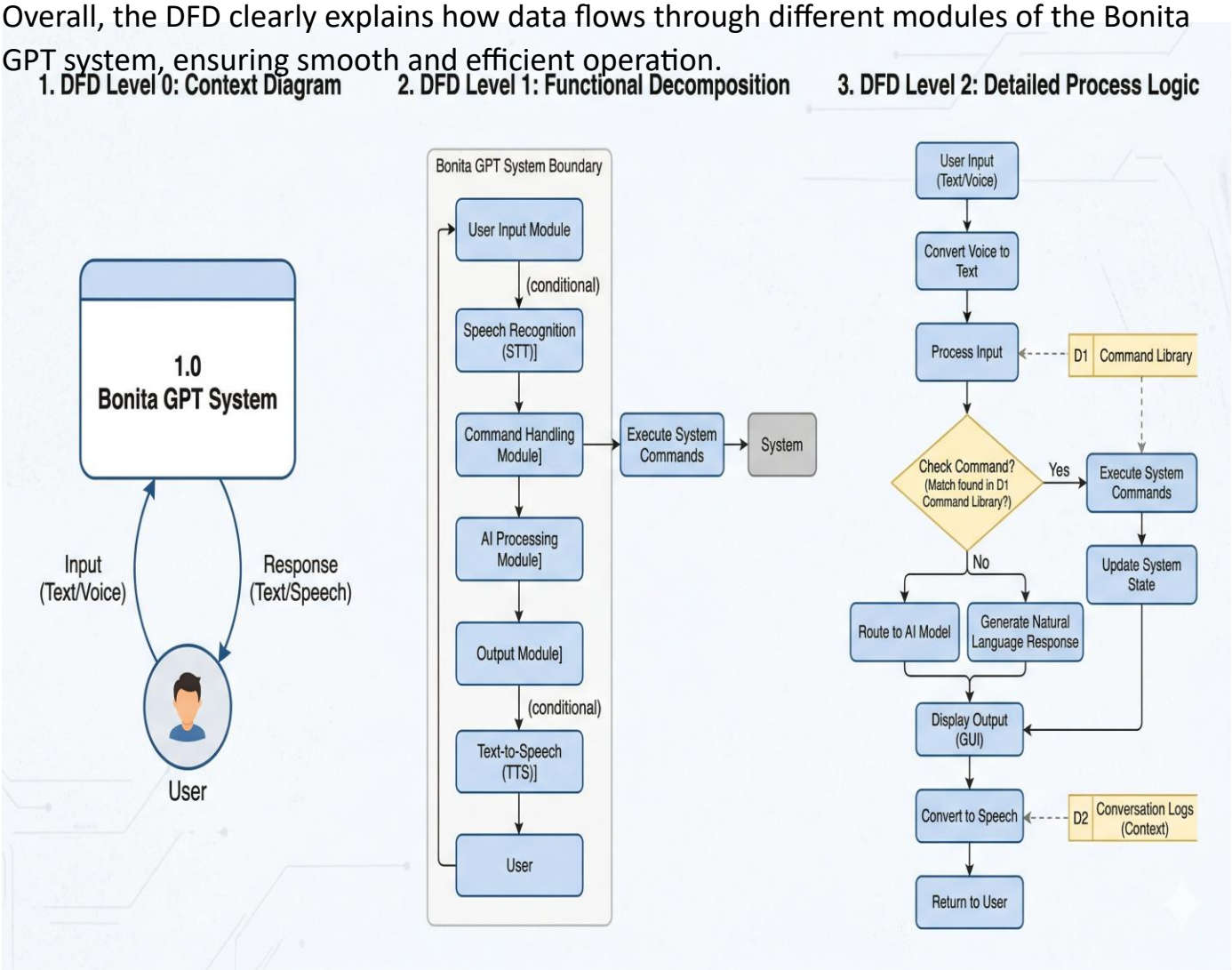
↓

Return to User

COMPONENTS OF DFD

- **External Entity:** User
- **Processes:** Input Handling, AI Processing, Command Execution, Output Generation
- **Data Flow:** User Input, Processed Text, Response Output
- **Data Store (Optional):** Command List / System Data

Overall, the DFD clearly explains how data flows through different modules of the Bonita GPT system, ensuring smooth and efficient operation.



ENTITY RELATIONSHIP DIAGRAM (ER DIAGRAM)

The Entity Relationship (ER) Diagram represents the structure of the system by showing the entities involved, their attributes, and the relationships between them.

MAIN ENTITIES

1. User

Represents the person interacting with the system.

Attributes:
- User_ID
- Input_Text
- Voice_Input

2. System (Bonita GPT)

Represents the main application that processes input and generates responses.

Attributes:
- System_ID
- Processed_Input
- Response

3. Command

Represents predefined commands handled by the system.

Attributes:
- Command_ID
- Command_Name
- Action

4. AI Model

Represents the AI component used to generate intelligent responses.

Attributes:
• Model_ID
• Query
• Generated_Response

5. Output

Represents the final result shown to the user.

Attributes:
• Output_ID
• Text_Output
• Voice_Output

RELATIONSHIPS

• User → interacts with → System
• System → checks → Command
• System → sends query to → AI Model
• AI Model → returns → Response
• System → generates → Output
• Output → delivered to → User

ER DIAGRAM (STRUCTURE)

[User]

|

| interacts

↓

[System] -------- checks --------> [Command]

|

| sends query

↓

[AI Model]

|

| generates

↓

[Output]

|

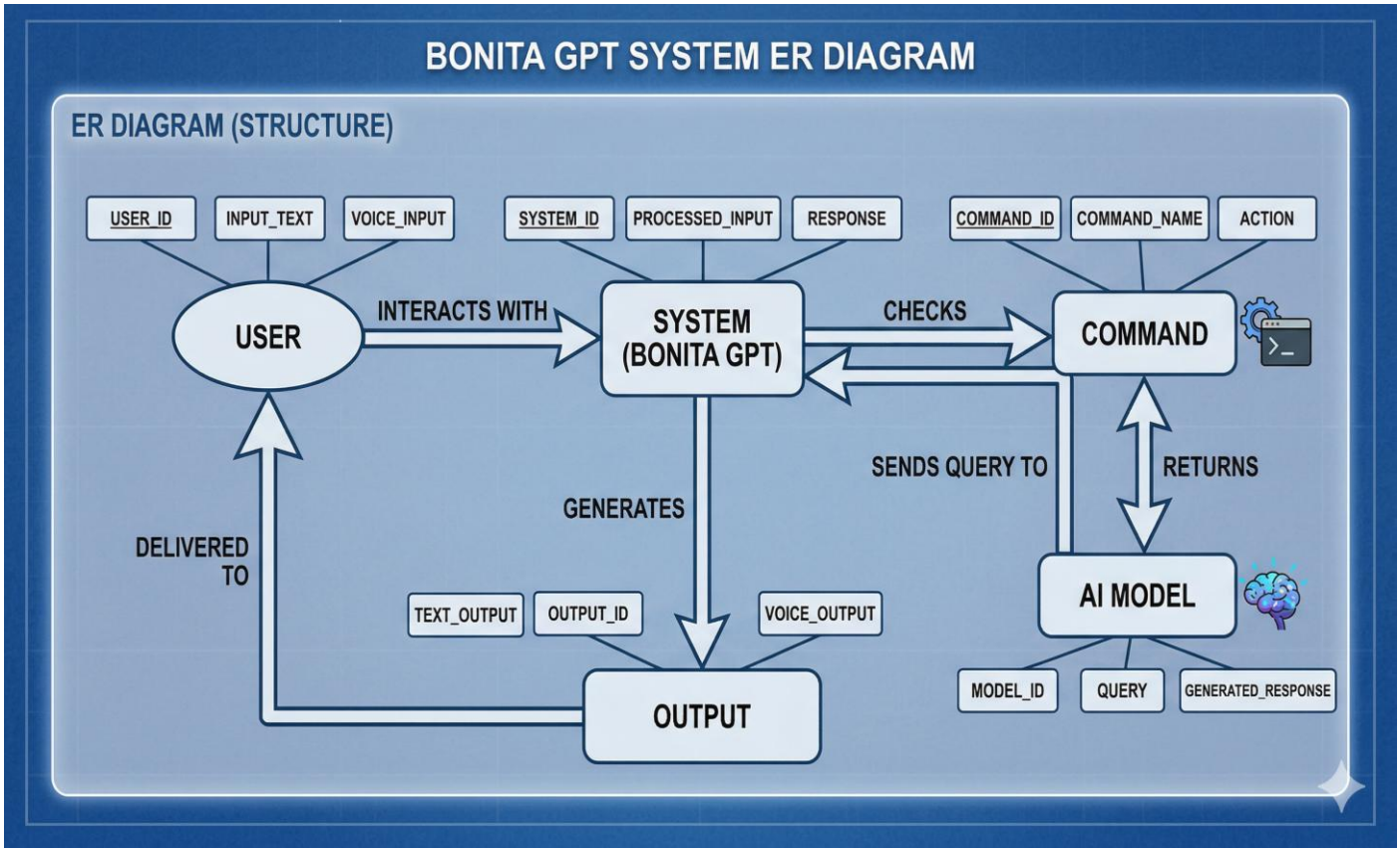
| delivered to

↓

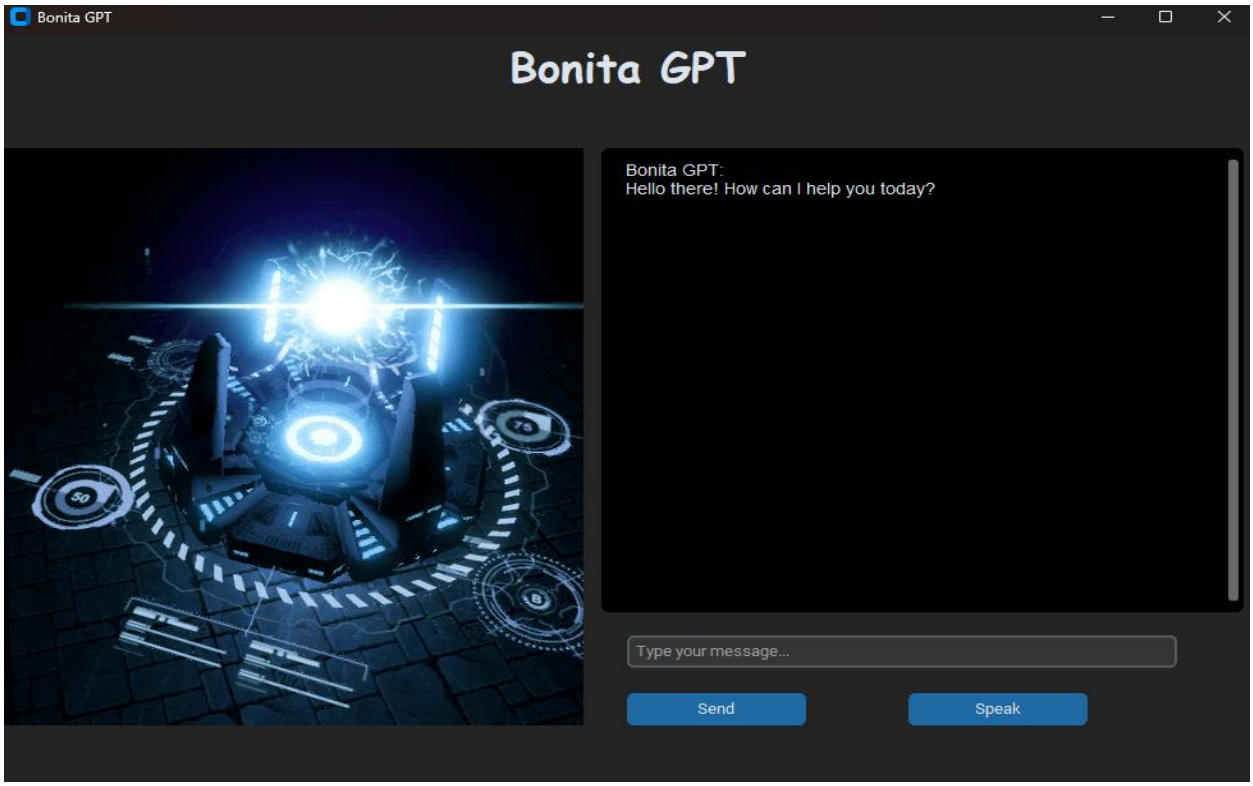
[User]

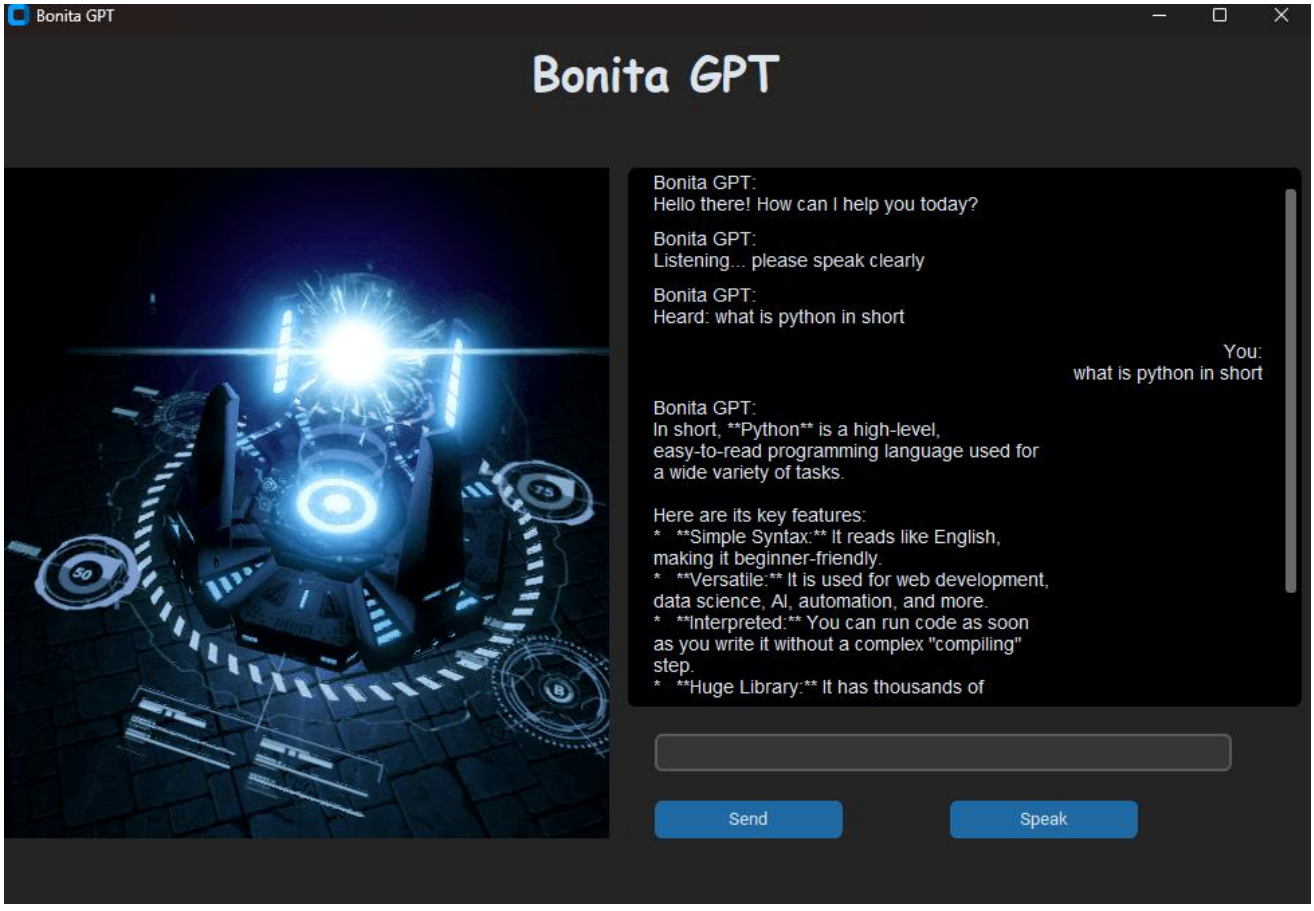
SUMMARY

The ER diagram shows how the user interacts with the system, how the system processes input using commands and AI, and how the final output is generated and delivered. It helps in understanding the logical structure and data relationships within the project.



Screenshots





ADVANTAGES

- **User-Friendly Interface:**

The system provides an easy-to-use graphical interface, making interaction simple and convenient for users.

- **Voice and Text Interaction:**

Supports both voice and text input, increasing accessibility and flexibility.

- **Real-Time Response:**

The assistant processes input quickly and provides instant responses.

- **Task Automation:**

Can perform basic tasks like opening websites, checking time, and executing commands, saving user effort.

- **AI-Based Interaction:**

Generates intelligent and context-aware responses, improving user experience.

- **Accessibility:**

Helpful for users who prefer voice interaction, including visually impaired users.

- **Modular Design:**

The system is easy to maintain and upgrade due to its modular structure.

DISADVANTAGES

- **Internet Dependency:**

AI response generation depends on internet connectivity for API access.

- **Limited Command Set:**

The system can only execute predefined commands and may not handle all user requests.

- **Speech Recognition Errors:**

Voice input may sometimes be inaccurate due to noise or pronunciation differences.

- **Performance Limitations:**

May slow down on low-end systems when handling multiple tasks.

- **Security Concerns:**

Lack of advanced authentication may lead to privacy issues.

- **Limited Understanding:**

The AI may not always fully understand complex or ambiguous queries.

FUTURE SCOPE

The *Bonita GPT: Voice-Enabled AI Virtual Assistant* has significant potential for future improvements and enhancements. With advancements in Artificial Intelligence and related technologies, the system can be further developed to provide more advanced features and better performance.

Some possible future enhancements are as follows:

- **Advanced Natural Language Understanding:**
The system can be improved to better understand complex queries, emotions, and user intent for more human-like conversations.

- **Multilingual Support:**
Future versions can support multiple languages, allowing users from different regions to interact with the assistant easily.

- **Integration with Smart Devices:**
The assistant can be connected with IoT devices to control smart home appliances such as lights, fans, and security systems.

- **Personalization Features:**
The system can store user preferences and provide personalized responses, reminders, and suggestions.

- **Offline AI Capabilities:**
Enhancing the system to work without internet dependency by using local AI models can improve reliability and privacy.

- **Mobile Application Development:**
The project can be extended into a mobile app for Android or iOS platforms to increase accessibility.

• **Enhanced GUI Design:**
The graphical interface can be further improved with modern designs, themes, and better animations for an enhanced user experience.

• **Integration with External Services:**
The assistant can be connected to services like email, calendar, weather updates, and news APIs for more functionality.

• **Security Improvements:**
User authentication and data protection mechanisms can be added to ensure privacy and secure interactions.

Overall, the future scope of the Bonita GPT project is vast, and with continuous development, it can evolve into a highly intelligent and widely usable virtual assistant system.

CONCLUSION

The *Bonita GPT: Voice-Enabled AI Virtual Assistant* project successfully demonstrates the practical implementation of Artificial Intelligence in developing an interactive and user-friendly system. The application integrates multiple technologies such as speech recognition, text-to-speech, AI-based response generation, and graphical user interface design to create a smart virtual assistant.

The system is capable of accepting both text and voice input, processing user queries in real time, and providing meaningful responses in both text and audio formats. It also performs basic tasks such as opening applications and providing real-time information like date and time, making it a useful tool for everyday use.

The modular design of the project ensures that each component functions efficiently and can be easily maintained or upgraded. This approach also makes the system scalable for future enhancements.

Through the development of this project, practical knowledge of Python programming, AI integration, GUI development, and voice processing technologies has been gained. It also highlights the growing importance of intelligent systems in improving human-computer interaction.

In conclusion, the project achieves its objectives and provides a strong foundation for building more advanced AI-based virtual assistants in the future.

REFERENCE

The following references were used during the development of the Bonita GPT: Voice-Enabled AI Virtual Assistant project:

• Python Software Foundation. (n.d.). Python Documentation. Retrieved from https://docs.python.org/3/

• Google. (n.d.). Google Generative AI Documentation. Retrieved from https://ai.google.dev/

• Zhang, A. (n.d.). SpeechRecognition Library Documentation. Retrieved from https://pypi.org/project/SpeechRecognition/

• pyttsx3 Contributors. (n.d.). pyttsx3 Documentation. Retrieved from https://pyttsx3.readthedocs.io/

• Schimansky, T. (n.d.). CustomTkinter Documentation. Retrieved from https://customtkinter.tomschimansky.com/

• Clark, A. (n.d.). Pillow (PIL) Documentation. Retrieved from https://pillow.readthedocs.io/

• W3Schools. (n.d.). Python Tutorial. Retrieved from https://www.w3schools.com/python/

• Stack Overflow Community. (n.d.). Stack Overflow. Retrieved from https://stackoverflow.com/

• GeeksforGeeks. (n.d.). Python Programming Articles. Retrieved from https://www.geeksforgeeks.org/

• YouTube. (n.d.). Programming Tutorials for Python, GUI, and AI Integration.